CASE 06 / OPEN SOURCE | THREE MERGED FIXES

Swift Reliability Fixes Kingfisher & Firefox iOS

Merged, reviewable fixes spanning concurrency, disk-cache behavior, and memory ownership.

Swift | iOS Development | Bug Fix | Performance Optimization | Unit Testing

PROBLEM

Mobile reliability problems can begin with a small implementation detail: a collection accessed concurrently, an unnecessary filesystem operation, or a closure retaining its owner. I contributed targeted Swift fixes for these problems in Kingfisher and Firefox iOS, with public pull requests that make the diagnosis and implementation reviewable.

CONTRIBUTION

In Kingfisher #2539, I changed cancellation to snapshot callback tokens while holding the existing lock, then cancel outside it. This addresses concurrent collection access without holding a non-recursive lock across an operation that acquires it again. A regression test exercises concurrent callback registration and cancellation. In #2556, I removed a redundant file-existence lookup before reading disk-cache metadata and covered cache-miss and error behavior.

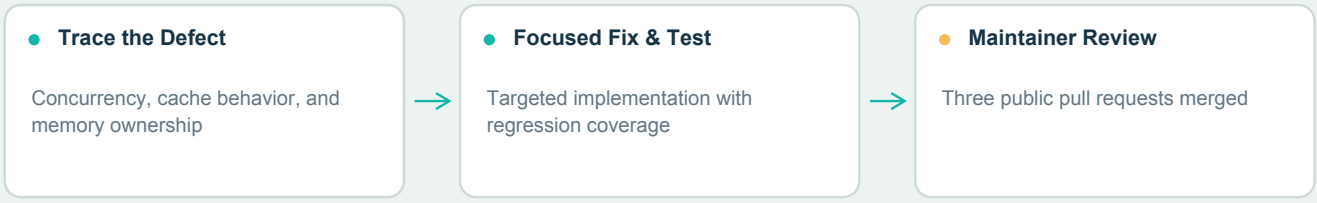
DECISIONS

The changes stayed close to each failure mechanism. Lock scope mattered as much as synchronization in the cancellation fix. The cache change reused the metadata path's missing-file handling. In Firefox iOS #34675, a setting-change closure retained the Settings controller through its settings collection; a weak capture broke that cycle. Its regression test populates the collection to reproduce the ownership structure.

RESULT

All three pull requests were merged, and both Kingfisher contributions were included in the 8.11.0 release. Together, they demonstrate focused debugging across concurrency, cache behavior, and memory lifetime, with regression coverage and maintainer review. The public implementation and review history let a prospective team inspect the reasoning behind the fixes when assessing work on an existing Swift codebase.

ENGINEERING FLOW



EVIDENCE

Kingfisher #2539 - merged cancellation data-race fix

Kingfisher #2556 - merged disk-cache metadata fix

Firefox iOS #34675 - merged retain-cycle fix

Kingfisher 8.11.0 - release notes